

CMPC132 Final Project Report

For this project, I made a number guessing game in Python called “Number Crunch.” The idea is pretty simple. The user picks a difficulty level, and the program generates a random number in a certain range. Then the user keeps guessing until they get it right. After each guess, the program tells them if they’re too high or too low.

I used functions to organize the code so it’s easier to read and reuse. The *number_generator()* function handles picking a random number depending on the difficulty. The *checker()* function compares the user’s guess to the actual number and prints feedback.

The main part of the program is a loop that keeps asking the user for guesses until they get it correct. I also added a counter to keep track of how many guesses it takes. To make sure the program doesn’t crash, I used a try/except block so if the user enters something that isn’t a number, it just asks them again instead of breaking.

One of the main issues was handling user input. People can type anything, so I had to make sure the program didn’t crash if the input wasn’t a number. Using try/except fixed that. Another small challenge was making sure the loop worked correctly and didn’t accidentally stop early or run forever. Using a boolean variable (*is_correct*) made that easier to control.

In this project, I learned a lot. It helped me get more comfortable with:

- Writing and using functions
- Loops and conditionals
- Handling errors with try/except
- Working with user input

It also helped me understand how to structure a simple program so it actually feels interactive. If I had more time, I would work on a few more things like:

- Add a limit on number of guesses
- Give better hints (like “you’re close”)
- Let the player restart without rerunning the program

Overall, the project works well and does what it's supposed to do. It's a simple game, but it shows the main programming concepts we learned in class. It also feels like something I could build on and improve later.